



School of Advanced Technology

Final Year Project

Project Specification Report

Project Title: Automated Morphological Subtype Classification of Organoids Using Machine Learning

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Student ID:

Project field: Bioimage Informatics

Supervisor:

Co-supervisor (if applicable):

1- Project Description and Problem Statement

Organoids are three-dimensional (3D) in vitro models that closely mimic the structure and function of human organs. They are widely used in drug screening and precision medicine. A critical feature is that different organoid morphological subtypes—such as “cystic” vs. “solid” in colorectal cancer, or “normal-like” vs. “dense” in oral cancer—correlate with distinct drug responses and patient prognoses [1, 2].

However, traditional analysis relies on manual visual inspection, which is subjective, slow, and poorly suited for large-scale experiments. Existing deep learning tools like OrganoID and OrgaExtractor can automatically segment and measure organoids, but they lack built-in subtype classification [3, 4].

To address this, this project aims to combine established segmentation tools with a custom machine learning classifier to develop an end-to-end pipeline for automated organoid subtyping.

2- Aims and Objectives

The primary aim is to develop an automated pipeline for classifying organoid morphological subtypes from microscopy images.

The specific objectives are:

Objective 1: Implement or adapt an existing deep learning model (based on U-Net architectures from OrganoID/OrgaExtractor) for accurate single-organoid segmentation from brightfield images.

Objective 2: Extract key morphological features (e.g., circularity, solidity, area, eccentricity) from segmented organoids.

Objective 3: Develop and validate a machine learning classifier (using a morphology score or a simple model like logistic regression) to classify organoids into known subtypes.

Objective 4: Evaluate the full pipeline's performance against manual annotations, using metrics like accuracy and Dice Similarity Coefficient (DSC).

3- Project Plan: Tasks and Milestones (Gantt Chart)

The project will be completed in two months (8 weeks). The plan focuses on rapid development and validation of a core prototype.

Week	Main Task	Key Activities	Milestone
1	Setup & Data Collection	Set up development environment (Python, PyTorch). Download a public organoid dataset (e.g., from OrganoID paper or other published sources).	M1: Dataset & Environment Ready
2	Model Implementation	Implement a simplified U-Net for segmentation (based on open-source code from OrganoID). Train on the collected dataset.	
3	Model Tuning	Fine-tune the segmentation model to achieve DSC > 0.7 on validation set.	M2: Functional Segmentation Model

Week	Main Task	Key Activities	Milestone
4	Feature & Classification	Code the feature extraction module. Implement a morphology scoring function (e.g., using linear equations from oral cancer study).	
5	Pipeline Integration	Combine segmentation + feature extraction + classification into an end-to-end script. Test on a small batch of images.	M3: End-to-End Pipeline Prototype
6	Evaluation	Evaluate the full pipeline on the test set. Report classification accuracy, segmentation DSC, and processing time.	M4: Evaluation Results Complete
7	Report Writing	Write the final project report covering methods, results, and discussion.	
8	Finalization	Clean code, add comments, prepare final demonstration slides.	M5: Final Report & Demo

4- Project Deliverables

At project completion, the following will be delivered:

Functional software pipeline (Python source code) for organoid segmentation, feature extraction, and subtype classification.

Trained model weights for the U-Net segmentation model.

Processed dataset (images, segmentation masks, and classification labels).

Final project report documenting methodology, evaluation results, and conclusions.

Final demonstration showing the pipeline processing new images and producing classification results.

5- Project Industrial Relevance

This project directly addresses a bottleneck in the pharmaceutical industry: the manual analysis of organoid images in high-throughput drug screening. The developed automated pipeline would allow companies to:

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- Accelerate drug screening: Quickly identify morphological changes induced by drug compounds.
 - Standardize quality control: Provide objective metrics for assessing organoid health across batches.
 - Enable larger studies: Scale up analysis from hundreds to thousands of images without human effort.

This capability is highly relevant to companies working on patient-derived organoids (PDOs) for personalized medicine.

References

(The references will follow an IEEE style as shown in the examples.)

Journals

[1] K. Huang, M. Li, Q. Li, Z. Chen, Y. Zhang, and Z. Gu, “Image-based profiling and deep learning reveal morphological heterogeneity of colorectal cancer organoids,” *Computers in Biology and Medicine*, vol. 173, p. 108322, May 2024.

[2] M. R. Lee et al., “Organoid morphology-guided classification for oral cancer reveals prognosis,” *Cell Reports Medicine*, vol. 6, no. 3, p. 102129, Mar. 2025.

[3] J. Matthews et al., “OrganoidID: A versatile deep learning platform for tracking and analysis of single-organoid dynamics,” *PLOS Computational Biology*, vol. 18, no. 11, p. e1010584, Nov. 2022.

[4] T. Park et al., “Development of a deep learning based image processing tool for enhanced organoid analysis,” *Scientific Reports*, vol. 13, no. 1, p. 22271, Dec. 2023.